

SHREYA RAGHURAJ

B.TECHSTUDENTATPESUNIVERSITY

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TECHNICAL SKILLS

C PROGRAMMING
PYTHON
HTML
JAVASCRIPT

MACHINE LEARNING
ARMSIM
MATLAB

LOGISIM
CHEMSKETCH
CISCO PACKET TRACER
RISC-V

RESEARCH/PROJECTS:

Ships SAR Image Analysis

October,2024

- Enhanced preprocessing revealed ship structures clearly, and multi-scale filtering reduced noise. The ResNet50 classifier reliably confirmed ship presence, creating a robust detection pipeline.
- Deep Learning Classification: A fine-tuned ResNet50 CNN distinguished ships from background. Data augmentation (rotations, flips, zooms) improved generalization. The model achieved 100% validation accuracy in three epochs, showing strong capability to identify ships in unseen images.
- Search & Rescue: Locating ships in distress under poor visibility.

Hospital management system (priority queue)

December,2025

- Built a terminal-based hospital triage system in C implementing a min-heap priority queue to manage patient treatment order based on emergency level.
- enqueue (heapify-up) and dequeue (heapify-down) — achieving $O(\log n)$ insertion and removal, ensuring critical patients are always served first

Libraray management system

December,2025

- Built with React.js, covering book cataloging, member management, and real-time availability tracking
- Implemented issuance, returns, overdue detection, and automatic fine calculation with a dashboard
- Dynamic filtering and search across genres, availability, and keywords

Tic-Tac-Toe

March,2026

- Board is a 3x3 char array, cells start as '1'-'9' and get replaced with 'X' or 'O' as players move
- Win detection checks all 3 rows, 3 columns, and 2 diagonals after every move — if all 3 match, that player wins
- two modes- AI (unbeatable) and human mode
- Input validation catches three bad inputs: non-numbers, out-of-range numbers, and cells already taken — loops until you give a valid move

EDUCATION

PSBB LEARNING LEADERSHIP ACADEMY

2011-2022

ACTIVITIES:

dancing , painting , public speaking , presentation.

BASE PU COLLEGE

2022-2024

GRADE: 11th-12th

science

PES UNIVERSITY - EC CAMPUS

2024-2028

Bachelor of Technology- BTech

Computer Science (Artificial Intelligence and machine learning)

ADDITIONAL INFORMATION

- Languages:** C, Python , HTML , JAVASCRIPT
- Certifications:** Introduction to programming using python(HackerRank)
- Awards/Activities:** model in Panache (PESU fashion club) , codefusion(CTF + hackathon)
- libraries:** pandas